Chapter 12: **Matrices and Graphs**

There are many connections between matrices and graphs. We consider several here: the powers of the adjacency matrix, cages, counting perfect matchings, and properties of the eigenvalues of a graph.

12.1 The Adjacency Matrix and Counting Walks

The ***adjacency matrix*** A of a graph with n vertices is the $n \times n$ matrix with entry $a_{ij} = 1$ if vertex i and j are adjacent and 0 otherwise.

Recall that in a walk one can repeat vertices and edges. Number the vertices from 1 up to n, and let $w_k(i,j)$ be the number of walks from i to j of length k. For example, the number of walks from i to i of length 2 is the degree of i.

THE GOODWILL–HUNTING THEOREM. *The value $w_k(i,j)$ equals the (i,j)-entry of A^k, where A is the adjacency matrix.*

PROOF. We can prove this by induction on k. The base case is $k = 1$, which is true, since the adjacency matrix A^1 by definition gives the number of length-1 walks. In general, assume we know it is true for k and test for $k+1$. Each walk from i to j of length $k+1$ can be split into a walk of length k from i to some vertex ℓ followed by the edge ℓ to j. So the total number is

$$w_k(i,j) = \sum_{\ell=1}^{n} w_k(i,\ell) w_1(\ell,j) = \sum_{\ell=1}^{n} A^k_{i,\ell} A_{\ell,j} = (A^k A)_{i,j} = A^{k+1}_{i,j},$$

as required. ◻

12.2 Cages

The next result is about the existence of particular graphs. Consider an r-regular graph with girth 5 (no triangle nor 4-cycle). Then a vertex, its neighbors, and their neighbors must all be distinct. This means there must be at least $1 + r + r(r-1) = r^2 + 1$ vertices. The question is, when is there such a graph with exactly $r^2 + 1$ vertices? Such a graph is called a ***cage***

For example, when $r = 2$ one has C_5, and when $r = 3$ one has the Petersen graph. Using matrix algebra, we can almost solve this question.

THEOREM. *If there is an r-regular graph on $r^2 + 1$ vertices with girth 5, then r must be one of 2, 3, 7, or 57.*

PROOF. Let G be such a graph. Then the conditions mean that the diameter of G is 2. In fact, every two vertices are either adjacent or have a unique common neighbor. That is, if i and j are adjacent there is no walk of length 2 between them, and if i and j are not adjacent there is exactly one walk of length between them.

So by the above theorem this means that the matrix $A^2 + A$ has r's on the diagonal and 1's everywhere else. As a matrix equation we have:

$$A^2 + A - (r-1)I = J,$$

where J is the all-1 matrix.

Now let λ be an eigenvalue of A with eigenvector $\boldsymbol{v}$. The above equation implies that $\boldsymbol{v}$ is an eigenvector of J with eigenvalue $\lambda^2 + \lambda - (r-1)$. A common exercise in a linear algebra class is to determine the eigenspaces of J. It can be checked that the matrix J has eigenvalue n with multiplicity 1 and the all-1-vector as eigenvector; and eigenvalue 0 with multiplicity $n - 1$.

The all-1 vector is an eigenvector of A with eigenvalue r. The above discussion means all other eigenvalues of A satisfy $\lambda^2 + \lambda - (r-1) = 0$. This equation has roots

$$\lambda^+ = \frac{-1 + \sqrt{1 + 4(r-1)}}{2} \quad \text{and} \quad \lambda^- = \frac{-1 - \sqrt{1 + 4(r-1)}}{2}.$$

Now, suppose λ^+ has multiplicity b as eigenvalue of A. It follows that λ^- has multiplicity $n - 1 - b$. Recall from linear algebra that the sum of the eigenvalues with multiplicities is the trace, which for A is clearly 0. So we have that

$$b\lambda^+ + (r^2 - b)\lambda^- + r = 0.$$

Thus we have an equation for b, but b must be an integer. Some algebra later we get that r must be one of the values given in the statement of the theorem. ⌑

There is a graph known for the case $r = 7$. But the case $r = 57$ remains unresolved.

12.3 Perfect Matchings, Determinants and Permanents

Recall that the determinant of a matrix is given by a formula that involves the sum over permutations of the product of entries in the matrix:

$$\det A = \sum_{\pi} (-1)^{\operatorname{sgn}(\pi)} \prod_i a_{i\pi(i)}$$

The ***permanent*** of A has the same formula except that the sign is deleted (all terms are added).

Theorem. *For a graph G, the permanent of A has the same parity as the number of perfect matchings of G.*

Proof. Each term $\prod_i a_{i\pi(i)}$ corresponds to a directed graph G_π where each vertex has in- and out-degree one and each arc is an orientation of some edge in G. Thus G_π is a collection of disjoint cycles, where some of the cycles have length 2. If G_π has a nontrivial cycle, then one can reverse the first such cycle to produce another graph. Thus we can pair off the π where G_π has a nontrivial cycle. We are left with the G_π that have only trivial cycles—but these correspond to a perfect matching. That is, the permanent of A has the same parity as the number of perfect matchings. ⌑

This led to Lovász's result:

Theorem. *A graph has an even number of perfect matchings if and only if there is a non-empty set S of vertices such that every vertex in the graph (S included) has an even number of neighbors in S.*

Proof. Consider a vector x and consider the product Ax. The i^{th} entry in the product is the sum of the values of x on the neighbors of the i^{th} vertex. It follows that if x is a 0-1-vector whose 1's correspond to S, that $Ax = 0$ in $\mathbb{Z}_2$ if and only if every vertex has an even number of neighbors in S. But by linear algebra, $Ax = 0$ has a nontrivial solution if and only if A has determinant zero. And working in $\mathbb{Z}_2$, the determinant and permanent are equal. And so the conclusion holds by the above theorem. ⌑

There are several other theorems that connect matchings to determinants.

12.4 Eigenvalues

The ***adjacency matrix*** A of a graph with n vertices is the $n \times n$ matrix with entry $a_{ij} = 1$ if vertex i and j are adjacent and 0 otherwise. The ***eigenvalues*** of the graph are the eigenvalues of the adjacency matrix. That is, $Ax = \lambda x$ for some nonzero vector x. By linear algebra, the eigenvalues are the roots of the characteristic polynomial, which is the determinant $\det(\lambda I - A)$. We deal only with the undirected graph, so the matrix is symmetric. Because the matrix is symmetric, the eigenvalues are real and there is a full set of eigenvectors.

If one views the vector x as weights on the vertices, then the i^{th} entry in Ax is the sum of the weights of the neighbors of the i^{th} vertex. So, for an eigenvector, one needs the sum of the weights of neighbors to be λ times the vertex's weight, for all vertices.

Example. *Consider the graph K_3. The adjacency matrix has 0's on the diagonal and 1's everywhere else. Thus*

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -1 & \lambda \end{vmatrix} = \lambda^3 - 3\lambda - 2.$$

The characteristic polynomial has roots 2 and -1 twice. If we put 1 on each vertex, we get eigenvector for $\lambda = 2$. If we use the three weights $\{0, 1, -1\}$ in any order, we have an eigenvector for $\lambda = -1$.

12.5 Regular Graphs and Bipartite Graphs

Example. *For the complete graph K_n the pattern for K_3 generalizes. If we put 1 on each vertex, we get an eigenvector for $\lambda = n - 1$. If we use weights $\{1, -1\}$ on two vertices and 0 elsewhere, we have an eigenvector for $\lambda = -1$. One can choose $n - 1$ linearly independent such vectors. (Why?) Thus the eigenvalues of K_n are $n - 1$ once and -1 $n - 1$ times.*

The eigenvector generalizes for regular graphs:

FACT. *If graph G is r-regular, then the all-1 vector is eigenvector for $\lambda = r$.*

EXAMPLE. *The complete bipartite graph $K_{m,n}$. Let's try for an eigenvector where the weights are constant on each partite set; say the m vertices have weight a, and the n vertices have weight b. Then the requirement is that $\lambda a = nb$ and $\lambda b = ma$. Multiplying these together, we get $\lambda^2 ab = abmn$, so that $\lambda = \pm\sqrt{mn}$ is an eigenvalue. Okay: we must check that we can choose a and b nonzero, but that is easily checked. Also, 0 is an eigenvalue: put 1 and -1 on one vertex of the same partite set, and 0's elsewhere. One can do this $m+n-2$ linearly independent ways (why?). Thus the eigenvalues of $K_{m,n}$ are $\sqrt{mn}$, $-\sqrt{mn}$, and 0 $m+n-2$ times.*

This symmetry of the eigenvalues generalizes to all bipartite graphs:

FACT. *If graph G is bipartite, then the multiplicity of λ is the same as the multiplicity of $-\lambda$.*

The reason is that if ones takes an eigenvector for λ, and multiplies all weights in one partite set by -1, one gets an eigenvector for $-\lambda$.

12.6 Further Results

EXAMPLE. *It is known that the eigenvalues of C_n are $2\cos(2\pi r/n)$ for $0 \leq r < n$.*

THEOREM. *If graph G has eigenvalues $\{\lambda_i\}$ and graph H has eigenvalues $\{\mu_j\}$, then the cartesian product $G \Box H$ has eigenvalues $\{\lambda_i + \mu_j\}$.*

PROOF. Let λ be an eigenvalue of G with eigenvector x, and μ an eigenvalue of H with eigenvector y. Construct a weighting of $G \Box H$ by giving vertex (u_i, v_j) the weight $x_i \times y_j$. Then, the sum of the weights of the neighbors of vertex (u_i, v_j) is

$$\begin{aligned} \sum_{k \in N_G(u_i)} (x_k y_j) + \sum_{k \in N_H(v_j)} (x_i y_k) &= y_j \sum_{k \in N_G(u_i)} x_k + x_i \sum_{k \in N_H(v_j)} y_k \\ &= y_j \lambda x_i + x_i \mu y_j \\ &= x_i y_j (\lambda + \mu). \end{aligned}$$

Since x and y are both not the zero vector, this weighting is not all-zero. And so the result is an eigenvector for $\lambda + \mu$. Can then show that multiplicities carry over too. ⌑

For example, the eigenvalues of the hypercube Q_n are $n, n-2, n-4, \ldots, -n$ with multiplicities given by binomial coefficients.

One can also obtain some information about the characteristic polynomial. Written as the form $\phi = \lambda^n + c_1\lambda^{n-1} + \ldots + c_n$, it holds that $c_1 = 0$ and that c_2 is minus the number of edges. This comes from the determinant formula

$$\det H = \sum \left\{ (-1)^{\operatorname{sgn}(\pi)} h_{i\pi(i)} : \pi \text{ is a permutation} \right\}.$$

Note that a permutation corresponds to a directed graph where we draw an arrow from each i to $\pi(i)$. Thus we get a nonzero contribution only when π corresponds to a union of a matching and directed cycles. This permutation formula then yields a recurrence for the characteristic polynomial:

$$\phi_G = \lambda\phi_{G-v} - \sum_{u \in N(v)} \phi_{G-v-u} - 2 \sum_{\text{cycle } C} \phi_{G-C}.$$

This follows by considering the permutations were $\pi(v) = v$, where $\pi(\pi(v)) = v$, and other cases. The factor of 2 is from the fact that a cycle has two orientations. This can be used, for example, to obtain a recurrence for the characteristic polynomial of the path and hence the cycle.

12.7 The Maximum and Second-Largest Eigenvalue

FACT. *If Δ is the maximum degree, then for all eigenvalues $-\Delta \leq \lambda \leq \Delta$.*

For, consider the vertex with the maximum absolute weight in an eigenvector. The absolute sum of its neighbors' weight is then at most its degree times its absolute weight. Indeed by results in linear algebra

$$\lambda_{\max} = \max\{\, x^t A x : x \text{ is unit vector} \,\}.$$

There is also an expansion property. For a set S of vertices, let $[S, \bar{S}]$ denote the edges with one end in S. The number of such edges is bounded by a function of the second largest eigenvalue. We prove here only the result for a regular graph:

THEOREM. *If graph is k-regular with order n and second largest eigenvalue λ, then*

$$|[S, \bar{S}]| \geq (k - \lambda)\, |S|\, |\bar{S}| \,/\, n.$$

Note that if every edge leaving a vertex in S went to a random vertex, then the expected number of edges leaving S would be $k|S||\bar{S}|/n$.

PROOF. Let L be the Laplacian matrix $D - A$ where D is the diagonal matrix with the degrees of the vertices (here k). The Laplacian has smallest eigenvalue 0, and second smallest eigenvalue $\lambda - k$.

Now, consider $x^t L x$. Multiplying out we get that

$$x^t L x = x^t k I x - x^t A x = k \sum x_i^2 - 2 \sum_{i \sim j} x_i x_j = \sum_{i \sim j} (x_i - x_j)^2.$$

We will pick x such that it is orthogonal to the all-1 vector. Then, by a similar proof to the previous section, we have that

$$x^t L x \geq (k - \lambda) x^t x.$$

Indeed, set $x_i = s$ for $i \in S$ and $x_i = -(n - s)$ for $i \notin S$, where $s = |S|/n$. A few lines of calculation checks that such x is orthogonal to the all-1 vector, that $x^t x = s(n - s)n$, and that $\sum_{i \sim j} (x_i - x_j)^2 = |[S, \bar{S}]| n^2$. Putting this all together yields the result. ⌑